palimpsest

User manual

Contents

1. Installation and compiling

Not yet published to the Typst Universe — import by path:

```
#import "path/to/palimpsest/lib.typ": *
```

A full project (manuscript, tracked manuscript, response letter) compiles with two commands:

```
typst compile --features bundle --format bundle main.typ
typst compile --features bundle --format bundle --input mode=tracked main.typ
```

The first produces `manuscript.pdf` and `response.pdf`; the second produces `manuscript-tracked.pdf`. Marking functions — `add`, `del`, `rep`, `passage`, `set-revisions` — also work directly in a single ordinary file, with a plain `typst compile` / `typst compile --input mode=tracked`, no bundle involved — that's the form used throughout the next chapter.

2. Marking changes: `add`, `del`, `rep`, `suppress`

`passage(anchors, body)` the citable unit — wraps `body` (plain text and/or `add`/`del`/`rep`/`suppress`) and attaches the anchor(s) a response letter will resolve back to this location. `anchors` is `none`, one label, or an array of labels; called as `passage(body)` (no anchors) for a typo fix with no comment to answer.

`add(body)` marks `body` as newly added.

`del(body)` marks `body` as removed.

`rep(old, new)` a replacement in one call.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

#passage[
  Propensity scores were estimated by logistic regression
  #add[and their overlap was assessed graphically].
]

#passage[
  The positivity assumption #rep[was not discussed][is now assessed graphically].
]
```

Compiled once clean:

> Propensity scores were estimated by logistic regression and their overlap was assessed graphically.
>
> The positivity assumption is now assessed graphically.

Once more with `--input mode=tracked`:

> Propensity scores were estimated by logistic regression and their overlap was assessed graphically.
>
> The positivity assumption ~~was not discussed~~ is now assessed graphically.

Two things to notice: `add` shows nothing extra in clean mode — `body` is just there, as if it had always been. `del` shows **nothing at all** in clean mode — no `hide()`, no reserved space; a deletion that survives into the clean version isn't a deletion.

2.1. `suppress`

`suppress(note)` like `del`, but never shows the real content, in **either** mode — only `note`, centered and italic, and only in tracked mode.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

#passage[
  The method section stands as before.
  #suppress[An accompanying diagnostic plot has been removed: it duplicated Figure 2.]
  The next paragraph is unaffected.
]
```

The method section stands as before.

[An accompanying diagnostic plot has been removed: it duplicated Figure 2.]

The next paragraph is unaffected.

The obvious question is why this exists next to `del`, which already hides its content in clean mode. The difference shows up specifically in the **tracked** manuscript: `del` still renders the real removed content there — struck through, as the previous section demonstrated — and for a figure, table, or equation, that real content is a real numbered element, which some templates' own numbering machinery can mishandle (the previous chapter's equation, struck-through-that-isn't, is a milder version of the same class of problem). `suppress` never emits a real figure/table/equation at all — only `note` — so there's nothing for any template to miscount, at the cost of the tracked manuscript no longer showing what was actually there.

`suppress` takes no content to hide, only `note` — there's no parameter for the real content at all, not even an unused one. Since that content is never rendered by this function, in any mode, keeping it as an argument would only be a place for some future edit to start rendering it by accident, undoing the whole point.

If you want the removed material kept at hand in the source to reconsider later, comment it out beside the call rather than deleting it outright — a `//` line is never evaluated by Typst, so there's no risk of it resurfacing on its own:

```
// #figure(table(...), caption: [The old table.])
#suppressed(<r2-1>, [Table removed: see response.])
```

2.2. `set-revisions`: every option, one at a time

Call once, before any `passage`, to change how tracked marks look. Takes effect for everything **after** the call — a document can use different styles in different sections.

```
#set-revisions(
  style: "inline",          // "inline" | "bar" | "none"
  color: auto,               // auto, or a fixed color for every mark
  add-style: underline,      // body -> content
  del-style: (smart default),// body -> content
  highlight-passage: false,  // tint the whole passage's background
  show-anchor: true,         // show [R1-2] at the end of the passage
  del-numbering: "none",     // "none" | "keep"
)
```

2.2.1. `style`

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

*style: "inline" (default)*
#passage[This whole sentence is marked #add[with an inline addition] in the flow of text.]

#v(0.8em)
```

```
#set-revisions(style: "bar")
*style: "bar"*
#passage[This whole sentence is marked #add[with an addition] under the bar style instead.]

#v(0.8em)
#set-revisions(style: "none")
*style: "none"*
#passage[This whole sentence is marked #add[with an addition] but renders exactly like the clean
version — a layout sanity check.]
```

style: "inline" (default) This whole sentence is marked with an inline addition in the flow of text.

style: "bar"
This whole sentence is marked with an addition under the bar style instead.

style: "none" This whole sentence is marked with an addition but renders exactly like the clean version — a layout sanity check.

`"inline"` (the default) underlines additions and strikes deletions in the flow of text. `"bar"` keeps that same underline/strike on the marks themselves and adds a colored vertical bar to the left of the whole passage's block — best when a passage forms its own block, not text mid-paragraph. `"none"` turns underline/strike off entirely, rendering exactly like the clean version even in the tracked compile — a layout sanity check, to confirm marking hasn't shifted anything.

2.2.2. `color`

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

*color: auto (default) — one color per reviewer*
#passage(<r1-1>)[Reviewer 1's #add[addition].]
#passage(<r2-1>)[Reviewer 2's #add[addition] — a different color, automatically.]

#v(0.8em)
#set-revisions(color: rgb("#7a7a7a"))
*color: a fixed value — overrides per-reviewer colors*
#passage(<r1-1>)[Reviewer 1's #add[addition], now in the fixed color.]
#passage(<r2-1>)[Reviewer 2's #add[addition], the same fixed color, not a different one.]
```

color: auto (default) — one color per reviewer Reviewer 1's addition. [R1-1] Reviewer 2's addition — a different color, automatically. [R2-1]

color: a fixed value — overrides per-reviewer colors Reviewer 1's addition, now in the fixed color. [R1-1] Reviewer 2's addition, the same fixed color, not a different one. [R2-1]

`auto` (the default) colors each mark by its anchor — one color per reviewer, a separate palette per co-author (a later chapter). A fixed color — `rgb(...)`, a named color — overrides that entirely: every mark gets the same color, regardless of anchor.

2.2.3. `add-style` / `del-style`

```
#import "../../lib.typ": *
```

```
#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

*add-style: underline, del-style: default (strikes plain text like this)*
#passage[Some text with #add[an addition] and #del[a deletion].]

#v(0.8em)
#let boxed-mark(body) = box(stroke: 0.6pt, inset: (x: 2pt), radius: 1.5pt, body)
#set-revisions(add-style: boxed-mark, del-style: boxed-mark)
*add-style/del-style: a custom `body -> content` function*
#passage[Some text with #add[an addition] and #del[a deletion], both boxed instead of underlined/
struck. The reviewer color still applies on top --- it's added separately by the passage, not by
this function.]
```

add-style: underline, del-style: default (strikes plain text like this) Some text with an addition and ~~a deletion~~.

add-style/del-style: a custom `body -> content` function Some text with $\boxed{\text{an addition}}$ and $\boxed{\text{a deletion}}$, both boxed instead of underlined/struck. The reviewer color still applies on top — it's added separately by the passage, not by this function.

Both take a single `body -> content` function — **not** `(body, color)`: the color is applied afterward, wrapped around whatever this function returns, so a custom style only needs to decide the visual treatment (box, highlight, …), never the color itself.

`add-style`'s default is plain `underline`. `del-style`'s default is smarter, because `strike()` never decorates a real math glyph or a figure's drawing — only literal text: ordinary text still gets struck, but a **block** equation, a figure, or a table gets a diagonal cross instead (figures/tables also keep their caption struck natively, since a caption is real text), and an **inline** equation gets a line straight through it rather than a full cross. A custom function passed here always replaces this default outright for every kind of content, the same way it would replace plain `strike`.

Deletions are also automatically desaturated relative to additions — a muted, darker version of the same reviewer's color, not merely "the same color with a different decoration". This matters specifically because the strike/cross/line above is the **only** other signal on math or on a figure's drawing: without a color difference, an added and a removed equation would otherwise look identical at a glance. This isn't a `del-style` option — it always applies, on top of whatever `del-style` renders.

2.2.4. `highlight-passage`

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

*highlight-passage: false (default)*
#passage[Only the #add[actually changed words] are tinted --- the rest of the sentence stays plain
black text.]

#v(0.8em)
#set-revisions(highlight-passage: true)
*highlight-passage: true*
#passage[The whole passage's background is lightly tinted, not just #add[the changed words] ---
useful when a reviewer asked for a full rewrite.]
```

highlight-passage: false (default) Only the actually changed words are tinted — the rest of the sentence stays plain black text.

highlight-passage: true
The whole passage's background is lightly tinted, not just the changed words — useful when a reviewer asked for a full rewrite.

`false` (the default) tints only the marks themselves — a passage where one word changed doesn't light up entirely. `true` tints the whole passage's background lightly, useful when a reviewer asked for a full rewrite and marking only the changed words would understate how much moved.

2.2.5. `show-anchor`

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

*show-anchor: true (default)*
#passage(<r1-2>)[The #add[anchor tag] appears in superscript at the end of the passage.]

#v(0.8em)
#set-revisions(show-anchor: false)
*show-anchor: false*
#passage(<r1-2>)[No tag at the end of #add[the passage] anymore --- still colored by reviewer,
just no visible label.]
```

show-anchor: true (default) The anchor tag appears in superscript at the end of the passage. [R1-2]

show-anchor: false No tag at the end of the passage anymore — still colored by reviewer, just no visible label.

The `[R1-2]` superscript at the end of a marked passage — what reviewers use most in practice to find their own comment in the manuscript without cross-referencing the letter. `false` removes the tag; the passage is still colored.

2.2.6. `del-numbering`

A figure, table, or equation that gets deleted still exists as a real element in the tracked manuscript — and by default would still consume a number, shifting every figure that comes after it out of sync with the clean version. `del-numbering: "none"` (the default) freezes the deleted element's number instead, so the **next** real figure keeps the same number in both versions:

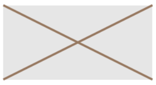
```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set math.equation(numbering: "(1)")
#set-revisions(require-exchange: false)

*del-numbering: "none" (default)*

#figure(rect(width: 2cm, height: 1cm, fill: luma(230)), caption: [Kept figure.]) <fig-a>
#passage(<x1>)[
  #del[#figure(rect(width: 2cm, height: 1cm, fill: luma(230)), caption: [Removed figure.]) <fig-
b>]
]
```

```
#figure(rect(width: 2cm, height: 1cm, fill: luma(230)), caption: [Next figure --- same number as
in the clean version.]) <fig-c>

#figure(table(columns: 2, [Kept], [table]), caption: [Kept table.]) <tab-a>
#passage(<x2>)[
  #del[#figure(table(columns: 2, [Removed], [table]), caption: [Removed table.]) <tab-b>]
]
#figure(table(columns: 2, [Next], [table]), caption: [Next table --- same number as in the clean
version.]) <tab-c>

Kept equation. $ a = b $ <eq-a-kept>
#passage(<x3>)[
  #del[$ E = m c^2 $ <eq-a>]
]
Next equation --- same number as in the clean version. $ c = d $ <eq-c-kept>
```

Tracked:

del-numbering: “none” (default)

Figure 1: Kept figure.



~~Removed figure.~~

[X1]

Figure 2: Next figure — same number as in the clean version.

Kept	table

Table 1: Kept table.

~~Removed~~	~~table~~

~~Removed table.~~

[X2]

Next	table

Table 2: Next table — same number as in the clean version.

Kept equation.

$$a = b \tag{1}$$

$$\cancel{E = mc^2}$$

[X3] Next equation — same number as in the clean version.

$$c = d \tag{2}$$

The **clean** compile of that same source, for comparison — the numbers match:

del-numbering: “none” (default)

Figure 1: Kept figure.

Figure 2: Next figure — same number as in the clean version.

Kept	table

Table 1: Kept table.

Next	table

Table 2: Next table — same number as in the clean version.

Kept equation.

$$a = b \quad (1)$$

Next equation — same number as in the clean version.

$$c = d \quad (2)$$

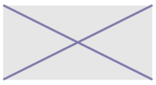
`"keep"` lets a deleted figure consume a number like anything else — rarely what you want (the whole point of `del-numbering: "none"` above is that a reviewer reading the tracked manuscript and citing “Figure 3” is citing the same Figure 3 that exists in the clean version you submit), but available:

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set math.equation(numbering: "(1)")
#set-revisions(require-exchange: false, del-numbering: "keep")

*del-numbering: "keep"*

#figure(rect(width: 2cm, height: 1cm, fill: luma(230)), caption: [Kept figure.]) <fig-d>
#passage(<x4>)[
  #del[#figure(rect(width: 2cm, height: 1cm, fill: luma(230)), caption: [Removed figure --- still
consumes a number here.]) <fig-e>]
]
#figure(rect(width: 2cm, height: 1cm, fill: luma(230)), caption: [Next figure --- number has
shifted, diverging from the clean version.]) <fig-f>

#figure(table(columns: 2, [Kept], [table]), caption: [Kept table.]) <tab-d>
#passage(<x5>)[
  #del[#figure(table(columns: 2, [Removed], [table]), caption: [Removed table --- still consumes a
number here.]) <tab-e>]
]
#figure(table(columns: 2, [Next], [table]), caption: [Next table --- number has shifted, diverging
from the clean version.]) <tab-f>

Kept equation. $ a = b $ <eq-a-kept-b>
#passage(<x6>)[
  #del[$ E = m c^2 $ <eq-b>]
]
Next equation --- number has shifted, diverging from the clean version. $ c = d $ <eq-c-kept-b>
```

Tracked:

del-numbering: “keep”

Figure 1: Kept figure.



~~Figure 2: Removed figure — still consumes a number here.~~

[X4]

Figure 3: Next figure — number has shifted, diverging from the clean version.

Kept	table

Table 1: Kept table.

~~Removed~~	~~table~~

~~Figure 4: Removed table — still consumes a number here.~~

[X5]

Next	table

Table 2: Next table — number has shifted, diverging from the clean version.

Kept equation.

$$a = b \tag{1}$$

$$\cancel{E = mc^2 \quad (2)}$$

[X6] Next equation — number has shifted, diverging from the clean version.

$$c = d \tag{3}$$

The **clean** compile of that same source — watch the third and fifth figures’ numbers diverge from the tracked version above:

del-numbering: “keep”

Figure 1: Kept figure.

Figure 2: Next figure — number has shifted, diverging from the clean version.

Kept	table

Table 1: Kept table.

Next	table

Table 2: Next table — number has shifted, diverging from the clean version.

Kept equation.

$$a = b \tag{1}$$

Next equation — number has shifted, diverging from the clean version.

$$c = d \tag{2}$$

Both examples also include a table and an equation, deleted the same way, worth looking at closely: the table’s cells and caption are struck through, same as ordinary text, and its diagonal cross covers the drawing itself. The equation gets the same diagonal cross rather than a strikethrough (see `del-style` above for why) — and, like the figure and the table, keeps its own number frozen under `del-numbering: "none"` so the equations around it stay in sync with the clean version, or consumes a real number under `"keep"`, exactly the same freeze/consume distinction shown above for figures and tables. A deleted equation still doesn’t need `suppress` for this reason alone; `suppress` (above) remains the answer for a template whose figure/table numbering ignores `del-numbering: "none"` outright (see the note further below), a problem specific to how some templates recompute figure/table numbers, not to equations.

2.3. Adding or removing a whole table row or column

`add`/`del` mark **content** — a word, a cell, a whole figure — but a table’s own shape is different: Typst’s `table()` has no built-in way to conditionally include or exclude an entire row or column depending on the compile mode. Getting this wrong doesn’t just look off — it can leave a row that reviewers were told was added missing from the manuscript actually submitted, or a removed row still sitting in it.

2.3.1. Adding a row or column

An added row or column stays in the table in **both** compiles, exactly like any other `#add[...]` — accepted new content never disappears from the clean version. No `mode()` check at all: wrap each cell in `add[...]` and include the row unconditionally, same as writing any other table row by hand.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

#passage[
  #figure(
    table(
      columns: 2,
      [Group], [N],
      [Control], [42],
      add[Treatment], add[45],
    ),
```

```
    caption: [Sample sizes by group.],
  )
]
```

Clean:

Group	N
Control	42
Treatment	45

Table 1: Sample sizes by group.

Tracked:

Group	N
Control	42
Treatment	45

Table 1: Sample sizes by group.

2.3.2. Removing a row or column

The opposite: a genuinely removed row or column must be **absent** from the clean compile, and there's no native way to hide part of a table conditionally. The fix is to build that row's cells as a spread array that's empty in clean and populated in tracked — `..if mode() == "clean" { () } else { (del[...], ...) }` — the same `mode()` the shortcuts chapter's `touched`/conditional-content cases already use, exported by `lib.typ` for exactly this.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

#passage[
  #figure(
    table(
      columns: 2,
      [Group], [N],
      [Control], [42],
      ..if mode() == "clean" { () } else { (del[Treatment], del[45]) },
      [Placebo], [40],
    ),
    caption: [Sample sizes by group.],
  )
]
```

Clean:

Group	N
Control	42
Placebo	40

Table 1: Sample sizes by group.

Tracked:

Group	N
Control	42
~~Treatment~~	~~45~~
Placebo	40

Table 1: Sample sizes by group.

2.3.3. Combining both in one table

A table can have an added row **and** a removed row at the same time — the two patterns above coexist without conflict, since neither changes `columns:`.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

#passage[
  #figure(
    table(
      columns: 2,
      [Group], [N],
      [Control], [42],
      ..if mode() == "clean" { () } else { (del[Placebo], del[40]) },
      add[Treatment], add[45],
    ),
    caption: [Sample sizes by group --- Placebo dropped, Treatment added.],
  )
]
```

Clean:

Group	N
Control	42
Treatment	45

Table 1: Sample sizes by group — Placebo dropped, Treatment added.

Tracked:

Group	N
Control	42
~~Placebo~~	~~40~~
Treatment	45

Table 1: Sample sizes by group — Placebo dropped, Treatment added.

Columns are the trap. `columns:` is one fixed count for the whole table, so an added column and a removed column do **not** belong in the same delta: an added column is part of the fixed base count (present in both compiles, like the row case above), and only a **removed** column belongs in a mode-dependent addition to that count — `base + int(mode() != "clean")` per removed column, never per added one. Writing it the other way around silently drops the added column from the clean, submitted manuscript — easy to miss, since `manuscript-tracked.pdf` still looks right.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

// Only the removed column enters the mode-dependent delta --- the
// added column is part of the fixed base count, present in both
// compiles, same as the added row above.
#passage[
  #figure(
    table(
      columns: 3 + int(mode() != "clean"),
      [Group], [N], add[p-value], ..if mode() == "clean" { () } else { (del[Old score],) },
      [Control], [42], add[0.03], ..if mode() == "clean" { () } else { (del[--],) },
    ),
    caption: [Sample sizes --- p-value column added, Old score column dropped.],
  )
]
```

Clean:

Group	N	p-value
Control	42	0.03

Table 1: Sample sizes — p-value column added, Old score column dropped.

Tracked:

Group	N	p-value	~~Old score~~
Control	42	0.03	–

Table 1: Sample sizes — p-value column added, Old score column dropped.

3. Shortcuts: `added`, `deleted`, `replaced`, `touched`, `suppressed`

`add`/`del`/`rep` mark part of a passage — a clause added mid-sentence, one phrase replacing another. When the **entire** passage is new, removed, or rewritten, wrapping it by hand (`passage(anchor)[#add[...]]`) is one level of nesting that never varies — these four shortcuts skip it.

`added(anchors, body, summary: none)` `passage(anchors, add(body))`.

`deleted(anchors, body, summary: none)` `passage(anchors, del(body))`.

`replaced(anchors, old, new, summary: none)` `passage(anchors, rep(old, new))`.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

#added(<r1-1>)[This whole sentence is new.]

#deleted(<r1-2>)[This whole sentence is being removed.]

#replaced(<r1-3>)[The old wording.][The new wording.]
```

[This whole sentence is new.] [R1-1]

~~This whole sentence is being removed.~~ [R1-2]

~~The old wording.~~ The new wording. [R1-3]

`summary:` is accepted by all three but only matters once a passage's tracked appearance gives a letter nothing worth quoting — `deleted` is the common case, covered below under `suppress`/`suppressed`.

3.1. `touched`

`touched(anchors, body)` declares a location without marking any change — "we checked this, it stays as written." Unlike `passage` on its own, an anchor with no `add`/`del`/`rep` inside doesn't need `allow-empty:` to avoid a warning; `touched` already sets it.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

#touched(<r1-4>)[This paragraph has not changed; we address the reviewer's concern in our response
instead.]
```

This paragraph has not changed; we address the reviewer's concern in our response instead. [R1-4]

Still colored and tagged like any other passage — just nothing struck or underlined, since nothing changed.

3.2. `suppressed`

`suppressed(anchors, note, summary: auto)` `passage(anchors, suppress(note), summary: ...)` — `suppress` (see the previous chapter) for a passage entirely made of it.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

#suppressed(<r2-1>, [Equation removed: superseded by the equation below.])

$ E = m c^2 (1 + epsilon) $

#suppressed(
  <r2-2>,
  [Table removed: see response to R2-2.], // shown in the tracked manuscript
  summary: [the old sensitivity table],   // for the letter, later
)
```

[Equation removed: superseded by the equation below.]

[R2-1]

$$E = mc^2(1 + \varepsilon)$$

[Table removed: see response to R2-2.]

[R2-2]

`note` and `summary:` are separate parameters on purpose. `note` has to read on its own, cold, in the middle of the manuscript — "Equation removed: …". `summary:` is inserted into a sentence the **letter** will already have written

— "Removed: ‹summary›" — so it should be a bare phrase, not repeat "removed" itself. `summary:` defaults to `note` when omitted, since most of the time the two don't need to differ; the second call above gives them separately.

3.3. Deleting a numbered figure under a template with its own numbering

`suppress`/`suppressed`, above, exist because of a real limitation: some templates (`charged-ieee`, `unequivocal-ams`) reimplement figure numbering with their own show rule, which doesn't cooperate with `del-numbering: "none"` — deleting a real `figure(...)` under one of these can leak a visible number onto the deleted figure, and even shift the numbers of every figure that follows it. `suppressed` sidesteps the whole problem by never emitting a real `figure()` at all — but it also never shows the original content, only a note.

When the deleted figure is short and worth keeping visible — not replaced by a note — skip `figure()` entirely instead: a plain block with the image and a hand-written caption never reaches the template's numbering machinery, because `del`'s automatic styling only special-cases a genuine `figure`/`table`/`math.equation`.

```
#import "../../lib.typ": *
#import "@preview/charged-ieee:0.1.3": ieee

// charged-ieee reimplements figure numbering with its own show rule,
// which doesn't cooperate with del-numbering: "none" -- a real
// figure() deleted here would leak a visible number and shift every
// figure that follows it (see CLAUDE.md, "del-numbering leak"). The
// workaround below never builds a figure() at all.
#show: ieee.with(
  title: [Probe],
  abstract: [none],
  authors: (
    (name: "A. Author", department: [], organization: [], location: [], email: []),
  ),
  index-terms: (),
  bibliography: none,
)

#set-revisions(require-exchange: false)

#figure(rect(width: 2.2cm, height: 1.3cm, fill: luma(200)), caption: [A figure, before.])

#deleted(<r1-1>, summary: [an obsolete diagnostic figure])[
  #align(center, block[
    #rect(width: 2.2cm, height: 1.3cm, fill: luma(200))
    #v(0.3em)
    #par(justify: false)[Fig. --- An obsolete diagnostic figure, kept visible.]
  ])
]

#figure(rect(width: 2.2cm, height: 1.3cm, fill: luma(200)), caption: [A figure, after.])
```

Clean:

Probe

A. Author

***Abstract*—none**

Fig. 1: A figure, before.

Fig. 2: A figure, after.

Tracked:

Probe

A. Author

***Abstract*—none**

Fig. 1: A figure, before.

[R1-1] ~~Fig. — An obsolete diagnostic figure, kept visible.~~

Fig. 2: A figure, after.

Figures 1 and 2 keep their real numbers on both sides — no leak, no shift. The trade-off: `del`'s automatic styling (desaturated color, strike) only decorates **text**, so the hand-written caption is struck and colored like any other deleted text, but the image itself isn't crossed out the way a genuinely deleted `figure()` is under a template with native numbering (previous chapter). Worth it whenever keeping the real content visible matters more than the missing cross mark; `examples/emoji-email/` uses this for exactly that reason, with an author note explaining why.

4. Anchors: reviewer, editor, author

An anchor is parsed into one of three kinds, purely from its shape: `<r1-2>` is reviewer 1, comment 2; `<e1>` is editor comment 1; anything else — `<bob-3>`, `<bob>` — is a co-author id, “bob”, optionally followed by a change number. Each kind gets its own color automatically, without any setup:

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(
  require-exchange: false,
  authors: (
    bob: (name: "Bobby Fischer", color: rgb("#c026d3")),
    alice: "Alice Smith",
  ),
)

#added(<r1-1>)[A reviewer's comment, `<r1-1>` --- one color per reviewer number.]

#added(<e1>)[An editor's comment, `<e1>` --- its own fixed color, distinct from every reviewer.]

#added(<bob-1>)[Bob's own change, `<bob-1>` --- name and color both registered explicitly.]

#added(<alice-1>)[Alice's own change, `<alice-1>` --- name registered, color assigned
automatically.]

#added(<carol-1>)[Carol's own change, `<carol-1>` --- nothing registered at all, still a distinct
color, displayed under her raw id.]
```

A reviewer’s comment, `<r1-1>` — one color per reviewer number. [R1-1]

An editor’s comment, `<e1>` — its own fixed color, distinct from every reviewer. [E1]

Bob’s own change, `<bob-1>` — name and color both registered explicitly. [BOB-1]

Alice’s own change, `<alice-1>` — name registered, color assigned automatically. [ALICE-1]

Carol’s own change, `<carol-1>` — nothing registered at all, still a distinct color, displayed under her raw id. [CAROL-1]

Reviewers are colored by number, cycling through a fixed palette. The editor gets one fixed color of its own. Co-authors draw from a **separate** palette, so a reviewer and a co-author active in the same manuscript are never confusable — and an author’s color holds even with nothing configured for them at all (Carol, above): it’s a deterministic function of the id text itself, the same color everywhere that id appears. `set-revisions(authors: (...))`, used above for Bob and Alice, additionally gives an id a full display name and/or a specific color:

```
#set-revisions(authors: (
  bob: (name: "Bobby Fischer", color: rgb("#c026d3")),  // both
  alice: "Alice Smith",                                   // name only
))
```

Either key can be omitted; an id with a `name:` but no `color:` still gets its automatic color, exactly like an id with nothing registered at all.

4.1. Multiple anchors on one passage

A comment raised jointly, or addressed in the same place — `passage`/`added`/`deleted`/`replaced` all accept an array of anchors instead of one:

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)
```

```
#added((<r1-2>, <r2-1>))[A concern raised jointly by two reviewers, addressed in one place.]
```

A concern raised jointly by two reviewers, addressed in one place. [R1-2, R2-1]

Colored by the first anchor in the list.

4.2. Bare anchors

`<bob>`, with no trailing `-<n>`, is deliberately **not** meant to key into one particular exchange — it's the shape for a project with no response document at all, where an anchor just says whose change this is:

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

#added(<dave>)[Dave's first change --- a bare anchor, no number.]

#added(<dave>)[Dave's second change, same bare anchor reused --- no "duplicate" warning, no "no
matching exchange" warning either, even though no response document exists anywhere in this
example.]
```

Dave's first change — a bare anchor, no number. [DAVE]

Dave's second change, same bare anchor reused — no "duplicate" warning, no "no matching exchange" warning either, even though no response document exists anywhere in this example. [DAVE]

Reused as many times as needed, with neither of the two diagnostics that would otherwise apply: no "duplicate exchange" (there's nothing to be a duplicate **of** — a bare anchor isn't meant to be unique), and no "no matching exchange" (covered next).

4.3. `require-exchange`

A **numbered** anchor is assumed to answer one specific comment, so `passage`/`added`/`deleted`/`replaced` check that a matching `exchange`/`note` exists somewhere and warn if not:

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

#added(<erin-1>)[A numbered anchor with no matching exchange anywhere --- warns by default.]

#set-revisions(require-exchange: false)

#added(<frank-1>)[Same situation, but `require-exchange: false` is now active --- no warning.]
```

⚠ **anchor erin-1: no matching exchange** A numbered anchor with no matching exchange anywhere — warns by default. [ERIN-1]

Same situation, but `require-exchange: false` is now active — no warning. [FRANK-1]

`set-revisions(require-exchange: false)` turns this check off from that point on — for a project that wants numbered, colored anchors without ever writing a response document.

5. Writing the exchanges: `reviewer`, `editor`, `exchange`

`reviewer(n, body)` groups a reviewer's comments under a heading colored by their number.

`editor(body)` same, for the editor's own comments.

`exchange(anchor, comment, response)` renders the quoted comment (in italics) and the response, with a header generated from the anchor.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

// Manuscript-side passages, matched by the exchanges below.
#added(<r1-1>)[A justification for the sample size.]
#added(<r1-2>)[A sensitivity analysis, added alongside it.]
#added(<e1>)[A clarified abstract word count.]

// Response-side exchanges -- a reviewer's block holds as many
// successive exchanges as needed, one heading for the whole group.
#reviewer(1)[
  #exchange(<r1-1>)[The sample size is not justified.][
    We agree and have added a justification.
  ]

  #exchange(<r1-2>)[A sensitivity analysis would strengthen this.][
    Added, using the same assumptions.
  ]
]

#editor[
  #exchange(<e1>)[Please double-check the abstract word count.][
    Confirmed within the journal's limit.
  ]
]
```

A justification for the sample size. [R1-1] A sensitivity analysis, added alongside it. [R1-2] A clarified abstract word count. [E1]

Reviewer 1

Reviewer 1 — comment 1

The sample size is not justified.

We agree and have added a justification.

Reviewer 1 — comment 2

A sensitivity analysis would strengthen this.

Added, using the same assumptions.

Editor

Editor — comment 1

Please double-check the abstract word count.

Confirmed within the journal's limit.

`exchange` checks, unconditionally, that its anchor matches a passage somewhere — an orphan comment answering nothing is always worth flagging, so this check has no `require-exchange`-style opt-out.

5.1. Co-authors: `author`, `note`

`author(id, body)` groups one co-author's own notes under a heading, colored and named the same way an anchor like `<bob-3>` already would be.

`note(anchor, text)` a single block — no comment to quote, just the author's own explanation. `exchange(anchor, text)`, with two arguments instead of three, renders identically.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(authors: (bob: "Bobby Fischer"))

// Manuscript-side passages.
#added(<bob-1>)[Bobby's own addition.]
#added(<bob-2>)[Bobby's second change.]

// Response-side notes -- no reviewer comment to quote, just an author
// explaining their own change.
#author("bob")[
  #note(<bob-1>)[Explaining why I made this change.]

  // exchange() with two arguments renders identically to note() above.
  #exchange(<bob-2>)[Explaining the second change, via exchange() instead.]
]
```

Bobby's own addition. [BOB-1] Bobby's second change. [BOB-2]

Bobby Fischer

Bobby Fischer — change 1

Explaining why I made this change.

Bobby Fischer — change 2

Explaining the second change, via exchange() instead.

5.2. `xcomment`

`xcomment(anchor)` a clickable cross-reference to another exchange, plus its page — "as already answered in comment R1-2."

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)

#added(<r1-1>)[The scope was narrowed to three use cases.]
#added(<r1-2>)[A sensitivity analysis was added, consistent with the narrower scope.]

#reviewer(1)[
  #exchange(<r1-1>)[The scope seems too broad.][
    Narrowed as suggested.
  ]

  #exchange(<r1-2>)[A sensitivity analysis would help.][
    Added --- see also our response to #xcomment(<r1-1>).
  ]
]
```

The scope was narrowed to three use cases. [R1-1] A sensitivity analysis was added, consistent with the narrower scope. [R1-2]

Reviewer 1

Reviewer 1 — comment 1
The scope seems too broad.

Narrowed as suggested.

Reviewer 1 — comment 2
A sensitivity analysis would help.

Added — see also our response to reviewer 1, comment 1, p. 1.

5.3. Header wording: `comment-word`, `change-word`, `term:`

The noun in a header — "comment" for reviewer/editor, "change" for a co-author — is `set-revisions(comment-word:, change-word:)`, global from that point on, or `term:` on one `exchange`/`note` call for just that occurrence:

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(authors: (bob: "Bobby Fischer"))

#added(<r1-1>)[Default word: "comment".]
#added(<bob-1>)[Default word: "change".]
#added(<r2-1>)[Global override applies here.]
#added(<bob-2>)[Per-call override applies here.]

#reviewer(1)[
  #exchange(<r1-1>)[A remark.][The default reviewer word.]
]
#author("bob")[
  #note(<bob-1>)[The default author word.]
]

#set-revisions(comment-word: "remark", change-word: "revision")

#reviewer(2)[
  #exchange(<r2-1>)[Another remark.][Now under the global override --- "remark", not "comment".]
]

#author("bob")[
  #note(<bob-2>, term: "aside")[This one overrides the word for just this call.]
]

Cross-references: #xcomment(<r1-1>), #xcomment(<r2-1>), and #xcomment(<bob-2>).
```

Default word: "comment". [R1-1] Default word: "change". [BOB-1] Global override applies here. [R2-1] Per-call override applies here. [BOB-2]

Reviewer 1

Reviewer 1 — comment 1
A remark.

The default reviewer word.

Bobby Fischer

Bobby Fischer — change 1
The default author word.

Reviewer 2

Reviewer 2 — remark 1
Another remark.

Now under the global override — "remark", not "comment".

Bobby Fischer

Bobby Fischer — aside 2
This one overrides the word for just this call.

Cross-references: reviewer 1, remark 1, p. 1, reviewer 2, remark 1, p. 1, and Bobby Fischer's aside 2, p. 1.

`xcomment` echoes whichever word the exchange it points to actually used — reading it back from that exchange's own data rather than recomputing it, so `#xcomment(<bob-2>)`'s explicit `term: "aside"` stays "aside" no matter where it's cited from. An **unoverridden** word, though, is resolved at the position where it's **read**, not where it was written: `<r1-1>`'s own exchange, near the top, renders "comment" — the default, in effect at that point — but `#xcomment(<r1-1>)` at the very bottom, after the global override, reads back "remark" for that same exchange. A global change to `comment-word`/`change-word` partway through a file affects every unoverridden reference read afterward, regardless of which word was showing where that exchange itself was originally written.

6. `pinpoint`: the manuscript/letter link

```
#pinpoint(anchor, excerpt: false, parens: true, verb: auto, show-page: true, quotes: false,
format: auto, mode: auto, on-empty: auto)
```

`pinpoint(<anchor>)` searches the manuscript for every passage carrying that anchor and reports where it is — the mechanism behind the real page numbers a response letter cites, since the manuscript and the letter share one bundle.

6.1. Page only (the default)

```
#import "../../lib.typ": *

#document("manuscript.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #passage(<r1-1>)[The sample size #rep[was not justified][was determined by a power
calculation].]
]

#document("response.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #reviewer(1)[
    #exchange(<r1-1>)[The sample size is not justified.][
      We thank the reviewer. #pinpoint(<r1-1>)
    ]
```

```
  ]
]
```

Manuscript:

The sample size was determined by a power calculation.

Response:

Reviewer 1

Reviewer 1 — comment 1

The sample size is not justified.

We thank the reviewer. (modified on p. 1)

The same anchor on two passages, on different pages, reports both:

```
#import "../../lib.typ": *

#document("manuscript.pdf")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #passage(<r1-1>)[This revision responds to a single comment that required changes in two
separate sections.]
  #pagebreak()
  #passage(<r1-1>)[The same comment, addressed a second time here.]
]

#document("response.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #reviewer(1)[
    #exchange(<r1-1>)[A single comment requiring changes in two places.][
      Addressed in both places. #pinpoint(<r1-1>)
    ]
  ]
]
```

Manuscript, page 1:

This revision responds to a single comment that required changes in two separate sections.

Manuscript, page 2:

The same comment, addressed a second time here.

Response:

Reviewer 1

Reviewer 1 — comment 1

A single comment requiring changes in two places.

Addressed in both places. (see p. 1 and p. 2)

If the anchor matches no real `add`/`del`/`rep` mark anywhere — a `touched` passage, cited only to point at text that didn't change — the verb switches from "modified on" to "see" automatically. Not a style choice: asserting a change that didn't happen would simply be false, so nothing has to be configured for it.

6.2. `parens:` and `verb:`: fitting the citation into a sentence

The wired-in parenthetical reads fine as a trailing citation ("We addressed this concern (modified on p. 3)."), but not every sentence ends that way:

```
#import "../../lib.typ": *

#document("manuscript.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #passage(<r1-1>)[The sample size #rep[was not justified][was determined by a power
calculation].]
  #touched(<r1-2>)[We kept the eligibility criteria exactly as submitted.]
]

#document("response.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #reviewer(1)[
    #exchange(<r1-1>)[The sample size is not justified.][
      See #pinpoint(<r1-1>, parens: false, verb: none) for the updated wording.
    ]

    #exchange(<r1-2>)[Please confirm the eligibility criteria are unchanged.][
      Confirmed --- #pinpoint(<r1-2>).
    ]
  ]
]
```

Manuscript:

The sample size was determined by a power calculation. We kept the eligibility criteria exactly as submitted.

Response:

Reviewer 1

Reviewer 1 — comment 1
The sample size is not justified.

See p. 1 for the updated wording.

Reviewer 1 — comment 2
Please confirm the eligibility criteria are unchanged.

Confirmed — (see p. 1).

`parens: false` drops the parentheses; `verb: none` additionally drops "modified on"/"see", leaving only "p. 3" (or "p. 3 and p. 7" for two pages) — for a sentence, like `See #pinpoint(<r>, parens: false, verb: none) for the updated wording.`, that already supplies its own verb. The two are independent: `parens: false` alone still says "modified on p. 3" without the parentheses; `verb: none` alone keeps the parentheses around a bare page number. Neither one alone fits every sentence shape — that's why both exist rather than a single "compact" switch.

6.3. `excerpt: true`: quoting the real text

Rather than a page number, the actual content of every passage carrying the anchor — exactly as it reads in the manuscript right now, so it can never go stale:

```
#import "../../lib.typ": *

#document("manuscript.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #passage(<r1-1>)[The sample size #rep[was not justified][was determined by a power
calculation].]
  #deleted(<r1-2>, summary: [an outdated caveat about generalizability])[
    An earlier caveat about generalizability, no longer needed.
  ]
]

#document("response.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #reviewer(1)[
    #exchange(<r1-1>)[The sample size is not justified.][
      Addressed as follows: #pinpoint(<r1-1>, excerpt: true)
    ]

    #exchange(<r1-2>)[This caveat is no longer accurate.][
      Agreed: #pinpoint(<r1-2>, excerpt: true)
    ]
  ]
]
```

Manuscript:

The sample size was determined by a power calculation.

Response:

Reviewer 1

Reviewer 1 — comment 1
The sample size is not justified.

Addressed as follows: **p. 1** — The sample size was determined by a power calculation.

Reviewer 1 — comment 2
This caveat is no longer accurate.

Agreed: Removed: an outdated caveat about generalizability.

A passage declared with `summary:` ignores `excerpt` and always renders "Removed: ‹summary›" instead, as the second comment above shows — there being no meaningful text left to quote in the clean manuscript once the passage is fully removed.

6.4. `show-page:` and `quotes:`: dropping the page, adding real quotation marks

```
#import "../../lib.typ": *

#document("manuscript.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #passage(<r1-1>)[The sample size #rep[was not justified][was determined by a power
calculation].]
]

#document("response.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #reviewer(1)[
    #exchange(<r1-1>)[The sample size is not justified.][
      #pinpoint(<r1-1>, excerpt: true, quotes: true)
```

```
      On page 1: #pinpoint(<r1-1>, excerpt: true, show-page: false)
    ]
  ]
]
```

Manuscript:

The sample size was determined by a power calculation.

Response:

Reviewer 1

Reviewer 1 — comment 1
The sample size is not justified.

p. 1 — "The sample size was determined by a power calculation."

On page 1: The sample size was determined by a power calculation.

`show-page: false` drops the leading `*p. 1* ---` — for a sentence that already states where the excerpt comes from ("On page 1, `<excerpt>`"), or a caller who simply doesn't want it. `quotes: true` wraps the excerpt in real, typeset quotation marks via Typst's native `quote()`.

`quotes: true` only ever **requests** quotation marks — a passage whose content is a figure, a table, or a block equation never gets them, regardless of this setting: forcing quotation marks onto a figure produces two stray quote glyphs sitting alone above and below it, not an improvement, so this is detected and declined automatically rather than left for you to remember passage by passage.

6.5. `mode:`: overriding style for one excerpt

An excerpt renders, by default, in whichever mode the **current** compile is running under — clean text only in `response.pdf` (the default with `letter: auto`, see letter:), struck-through/underlined tracked style in `response-tracked.pdf` (with `letter: true`). `mode:` overrides this for one call, in either direction.

`mode: "tracked"`, called from a clean letter, shows the tracked style even though `response.pdf` itself is otherwise all clean text:

```
#import "../../lib.typ": *

#document("manuscript.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #passage(<r1-1>)[The sample size #rep[was not justified][was determined by a power
calculation].]
]

#document("response.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #reviewer(1)[
    #exchange(<r1-1>)[The sample size is not justified.][
      Compare the wording: #pinpoint(<r1-1>, excerpt: true, mode: "tracked")
    ]
  ]
]
```

Manuscript:

The sample size was determined by a power calculation.

Response:

Reviewer 1

Reviewer 1 — comment 1

The sample size is not justified.

Compare the wording: **p. 1** — The sample size ~~was not justified~~ was determined by a power calculation.

Useful when the point being made is "we removed exactly what you objected to," which a clean, final-text quote doesn't convey on its own.

The reverse also works: `mode: "clean"`, called from a letter compiled tracked (`letter: true`, `--input mode=tracked`), shows the final wording for one excerpt even though the rest of that same letter is otherwise quoting tracked style:

```
#import "../../lib.typ": *

#let my-template(body) = {
  set page(width: 16.6cm, height: auto, margin: 12pt)
  set text(size: 10.5pt)
  body
}

#let exchanges = reviewer(1)[
  #exchange(<r1-1>)[This claim needs more support.][
    We have expanded this sentence. #pinpoint(<r1-1>, excerpt: true, mode: "clean")
  ]
]

#show: revisions.with(
  template: my-template,
  letter-template: my-template,
  exchanges: exchanges,
  letter: true,
)

#passage(<r1-1>)[The treatment #rep[has an effect][has a clinically meaningful effect] on
survival.]
```

Manuscript, tracked:

The treatment ~~has an effect~~ has a clinically meaningful effect on survival. [R1-1]

Response, tracked — the excerpt still reads as accepted, final text, despite the compile being tracked:

Reviewer 1

Reviewer 1 — comment 1

This claim needs more support.

We have expanded this sentence. **p. 1** — The treatment has a clinically meaningful effect on survival.

6.6. `on-empty:` and `format:`

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

No anchor `<r9-9>` exists anywhere in this document.

Default `on-empty:` warns: #pinpoint(<r9-9>)

#pinpoint(<r9-9>, on-empty: none) shows nothing instead when given `on-empty: none`.

#pinpoint(<r9-9>, on-empty: [not found in this round]) shows custom content instead, given as `on-
empty:`.

#passage(<r2-1>)[A change #add[on this page].]

Default wording: #pinpoint(<r2-1>)

Custom `format:`: #pinpoint(<r2-1>, format: (pages, has-marks) => [(see line 1 of p.
#pages.first())])
```

No anchor `<r9-9>` exists anywhere in this document.

Default `on-empty:` warns: ⚠ **pinpoint(<r9-9>): no revision attached to this anchor**

shows nothing instead when given `on-empty: none`.

not found in this round shows custom content instead, given as `on-empty:`.

A change on this page.

Default wording: (modified on p. 1)

Custom `format:`: (see line 1 of p. 1)

`on-empty:` controls what happens when no passage anywhere carries the given anchor — `auto` (the default) warns, `none` shows nothing, anything else is shown as-is. `format:` replaces the default page-list wording entirely, with a function `(pages-array, has-marks) -> content` — `has-marks` is the same flag that drives the automatic "modified"/"see" switch above, passed through so a fully custom format can make that same distinction without querying the bundle again. For a journal with its own citation convention.

7. `xref`: pointing into the manuscript

`xref(label)` like `@label`/`ref(label)` — already correct across the bundle, resolving to the manuscript's real number for free — but with the manuscript's real page number appended.

```
#import "../../lib.typ": *

#document("manuscript.pdf")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #set heading(numbering: "1.")
  = Results
  #pagebreak()
  #figure(rect(width: 3cm, height: 1.5cm, fill: luma(230)), caption: [Propensity score
distribution.]) <fig-a>
]

#document("response.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)

  A bare reference resolves the real manuscript number, for free: @fig-a.

  `xref` adds the page number on top: #xref(<fig-a>).
]
```

```
  A label that doesn't exist: #xref(<fig-nonexistent>).
]
```

Manuscript, page 2:

Figure 1: Propensity score distribution.

Response:

A bare reference resolves the real manuscript number, for free: Figure 1.

`xref` adds the page number on top: Figure 1, p. 2.

A label that doesn't exist: **⚠ xref(<fig-nonexistent>): not found**.

A bare `@fig-a` already gets the number right, since the letter and the manuscript share one bundle — `xref` only adds the page. Worth using explicitly anyway: a bare reference stays correct even if some future version of the package needs a second copy of the manuscript in the bundle for some other reason, where `@fig-a` alone would turn ambiguous between the two copies and `xref` would not. A label that doesn't exist anywhere warns rather than breaking the compile.

8. `change-list`: a summary table of every marked passage

`change-list(title: [Summary of changes], level: 1)` one row per marked passage — comment, type of change, page, section.

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set heading(numbering: "1.")
#set-revisions(require-exchange: false, authors: (bob: "Bobby Fischer"))

#change-list()

= Introduction

#added(<r1-1>)[A reviewer's addition.]
#added(<bob-1>)[Bobby's own addition.]

== Background

#added(<r1-2>)[A second reviewer comment, addressed in this subsection.]

= Methods

#added(<e1>)[An editor's addition.]
#added(none)[An anonymous typo fix, no anchor attached.]
#touched(<r9-9>)[This paragraph is unchanged --- never listed below.]
```

Summary of changes

Comment	Type	Page	Section
R1-1	addition	1	Introduction
R1-2	addition	1	Introduction
BOB-1	addition	1	Introduction
E1	addition	1	Methods
—	addition	1	Methods

1. Introduction

A reviewer's addition. [R1-1] Bobby's own addition. [BOB-1]

1.1. Background

A second reviewer comment, addressed in this subsection. [R1-2]

2. Methods

An editor's addition. [E1] An anonymous typo fix, no anchor attached. This paragraph is unchanged — never listed below. [R9-9]

Renders nothing in clean mode, like `del`/`suppress` — placed once anywhere in shared manuscript content, it only ever shows up in `manuscript-tracked.pdf`. Rows are sorted by comment identifier — reviewers in order, then co-authors alphabetically by id, then the editor, anchor-less passages last — so the table doubles as a checklist against the letter, not just a readout of document order. `touched` never appears: nothing changed there to list. A passage with several marks shows every kind it contains, joined with "/" (`addition/deletion`, say, for a manually mixed passage that isn't a single `rep`).

The "Section" column is the nearest preceding heading at `level:` (top-level by default) — notice the reviewer comment inside "Background" above is listed under "Introduction", its enclosing top-level section, not the subsection itself.

8.1. `level:`: which heading counts as "Section"

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set heading(numbering: "1.")
#set-revisions(require-exchange: false)

*level: 1 (default) --- "Section" names the top-level heading:*

#change-list()

*level: 2 --- "Section" names the subsection instead:*

#change-list(level: 2)

= Introduction

== Background

#added(<r1-1>)[A reviewer's addition, inside a subsection.]
```

level: 1 (default) — "Section" names the top-level heading:

Summary of changes

Comment	Type	Page	Section
R1-1	addition	1	Introduction

level: 2 — "Section" names the subsection instead:

Summary of changes

Comment	Type	Page	Section
R1-1	addition	1	Background

1. Introduction

1.1. Background

A reviewer's addition, inside a subsection. [R1-1]

Same passage, same position — only `level:` differs between the two calls, and the Section column changes with it: `1` (the default) names the enclosing top-level heading, `2` the enclosing subsection.

8.2. `title:`: avoiding a redundant heading

`change-list()` prints its own bold "Summary of changes" line above the table by default. If you already write your own heading right before the call, that becomes two headings back to back:

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

*Without `title: none` --- your own heading, then the package's own "Summary of changes" on top of
it:*

#text(weight: "bold", size: 1.2em)[Summary of Revisions]
#change-list()

*With `title: none` --- no second, redundant heading:*

#text(weight: "bold", size: 1.2em)[Summary of Revisions]
#change-list(title: none)

#added(<r1-1>)[A reviewer's addition.]
```

Without `title: none` — your own heading, then the package's own "Summary of changes" on top of it:

Summary of Revisions

Summary of changes

Comment	Type	Page	Section
R1-1	addition	1	—

With `title: none` — no second, redundant heading:

Summary of Revisions

Comment	Type	Page	Section
R1-1	addition	1	—

A reviewer's addition. [R1-1]

`title: none` removes the package's own line, leaving just your heading and the table — `title: [Some other text]` would replace it with different wording instead, for the same reason.

9. Bibliography

Typst 0.15′s multiple bibliographies handle three situations.

9.1. A citation inside `add`/`del`

Nothing special to do — it becomes part of the manuscript's own bibliography either way. The difference shows up in **which version** the citation appears in:

```
#import "../../lib.typ": *

#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

#added(<r1-1>)[A new claim, supported by @smith2020.]

#deleted(<r1-2>, summary: [an outdated claim])[
  An outdated claim, previously supported by @jones2019.
]

#bibliography("/docs/manual-snippets/biblio-manuscript.bib")
```

Clean — only the surviving citation:

A new claim, supported by [1].

Bibliography

[1] J. Smith, "A relevant finding," *Journal of Examples*, 2020.

Tracked — both, since the struck-through text is still there to resolve:

A new claim, supported by [1]. [R1-1]

~~An outdated claim, previously supported by [2].~~ [R1-2]

Bibliography

[1] J. Smith, "A relevant finding," *Journal of Examples*, 2020.

[2] A. Jones, "An outdated finding," *Journal of Examples*, 2019.

A citation inside a deleted passage disappears from the bibliography in the clean compile and reappears in the tracked one, so a reader following the struck-through text can still resolve what it cited — the opposite of `latexdiff`, notorious for leaving phantom, unresolvable references behind.

9.2. `letter-bibliography`: a citation the letter makes on its own

`letter-bibliography(path, title: [References cited in this response])` a second bibliography, scoped to citations made **inside** the letter and numbered independently from the manuscript's own.

A real project's three-file shape (`main.typ`/`manuscript.typ`/`responses.typ`, covered in the last chapter) — `manuscript.typ`:

```
#import "../../../../lib.typ": *

An earlier claim, supported by @jones2019.

#added(<r1-1>)[A second claim, supported by @smith2020.]

#bibliography("/docs/manual-snippets/biblio-manuscript.bib")
```

An earlier claim, supported by [1].

A second claim, supported by [2].

Bibliography

[1] A. Jones, "An outdated finding," *Journal of Examples*, 2019.

[2] J. Smith, "A relevant finding," *Journal of Examples*, 2020.

`responses.typ`:

```
#import "../../../../lib.typ": *

#reviewer(1)[
  #exchange(<r1-1>)[Please support this claim.][
    Done, see @smith2020. Also relevant, though not cited in the manuscript: @lee2022.
#pinpoint(<r1-1>)
  ]
]

For the reviewer's convenience only:

#figure(
  table(columns: 2, [Group], [N]),
  caption: [Sample sizes, not shown in the manuscript.],
)

#letter-bibliography("/docs/manual-snippets/biblio-letter.bib")
```

Response to Reviewers

Reviewer 1

Reviewer 1 — comment 1
Please support this claim.

Done, see [1]. Also relevant, though not cited in the manuscript: [2]. (modified on p. 1)

For the reviewer's convenience only:

Group	N

Table R1: Sample sizes, not shown in the manuscript.

References cited in this response

[1] J. Smith, "A relevant finding," *Journal of Examples*, 2020.

[2] G. Lee, "A related method, not cited in the manuscript," *Journal of Methods*, 2022.

`smith2020` is `[2]` in the manuscript's own bibliography but `[1]` in the letter's — two independent numbering sequences, not one shared list. This is also why `smith2020` had to be listed in **both** `.bib` files even though the manuscript already cites it: a citation made directly in the letter's own prose only ever resolves against the `.bib` passed to `letter-bibliography`, regardless of what the manuscript's own bibliography already knows.

Path gotcha: `path` must be root-relative (a leading `/`, resolved against `--root`), not relative to the file that calls `letter-bibliography`. Typst resolves a path string against the file that calls the path-consuming builtin — here, `bibliography()` inside `letter-bibliography`, in the package's own source — not the file that wrote the string literal. A plain `"responses.bib"` looks next to the package's own source and fails.

9.3. Letter numbering

A figure or table the letter adds **on its own** — "for the reviewer's convenience only," never in the manuscript, like the table above — is numbered independently from the manuscript's, prefixed `R`, so it can never be confused with a manuscript one.

A figure or table `pinpoint(excerpt: true)` re-emits **from** the manuscript is different: it keeps the manuscript's own real number, not an `R` one — "Figure 1" reads "Figure 1" in the letter too, not "Figure R1":

```
#import "../../lib.typ": *
#import "../../src/letter.typ": with-letter-numbering

#document("manuscript.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #passage(<r1-1>)[
    #add[
      #figure(rect(width: 3cm, height: 2cm, fill: luma(220)), caption: [A subgroup analysis, added per reviewer request.]) <fig-subgroup>
    ]
  ]
]

#document("response.png")[
  #set page(width: 16.6cm, height: auto, margin: 12pt)
  #set text(size: 10.5pt)
  #with-letter-numbering[
    #reviewer(1)[
      #exchange(<r1-1>)[Please add a subgroup analysis.][
        Done. #pinpoint(<r1-1>, excerpt: true)
      ]
    ]

    For the reviewer's convenience only, not in the manuscript:
```

```
    #figure(rect(width: 3cm, height: 2cm, fill: luma(150)), caption: [A figure the letter adds on
its own.])
  ]
]
```

Manuscript:

Figure 1: A subgroup analysis, added per reviewer request.

Response:

Reviewer 1

Reviewer 1 — comment 1
Please add a subgroup analysis.

Done. **p. 1** —

Figure 1: A subgroup analysis, added per reviewer request.

For the reviewer's convenience only, not in the manuscript:

Figure R2: A figure the letter adds on its own.

The letter's own figure above is captioned "Figure R2," not "Figure R1" — the excerpt before it still occupies the first slot of the letter's own counter, even though what it **displays** is the manuscript's number rather than that slot's. Already true before either was ever matched to the manuscript (an excerpt has always advanced the letter's counter); the only thing that changed is what a matched excerpt shows on the page.

10. Diagnostics and strict mode

Typst has no public API for a script to emit its own compiler warning, so every check below shows instead as a visible box, right at the fault — muted in the clean compile for anything embedded in the manuscript itself (`manuscript.pdf` must never carry one, since it's the file actually sent out under review), shown regardless of mode for anything embedded in the letter (`response.pdf` has no separate tracked compile to show it in instead).

Two already appeared earlier, in context: a numbered anchor with no matching exchange and a bare anchor's exemption from it; `pinpoint` with no matching anchor (`on-empty:`). Four more:

```
#import "../../lib.typ": *
```

```
#set page(width: 16.6cm, height: auto, margin: 12pt)
#set text(size: 10.5pt)
#set-revisions(require-exchange: false)

*Comment with no matching passage anywhere:*
#reviewer(1)[
  #exchange(<r9-1>)[A comment with nothing in the manuscript to answer.][A response anyway.]
]

*Exchange with a blank response:*
#passage(<r9-2>)[#add[A real addition, matched below.]]
#reviewer(1)[
  #exchange(<r9-2>)[A real comment.][]
]

*A mark outside any passage:*
#add[Added text with no enclosing `passage`.]

*A passage with no mark and no `summary:`:*
#passage(<r9-3>)[Plain text, nothing added, removed, or replaced.]
```

Comment with no matching passage anywhere:

Reviewer 1

⚠ comment r9-1 has no matching revision in the manuscript **Reviewer 9 — comment 1**

A comment with nothing in the manuscript to answer.

A response anyway.

Exchange with a blank response: A real addition, matched below. [R9-2]

Reviewer 1

⚠ exchange r9-2: empty response **Reviewer 9 — comment 2**

A real comment.

A mark outside any passage: ⚠ #add outside any passage: no anchor, no page Added text with no enclosing passage.

A passage with no mark and no `summary`: ⚠ passage r9-3: contains no mark (did you mean touched?)
Plain text, nothing added, removed, or replaced. [R9-3]

Four more exist but aren't demonstrated live here, since each needs a slightly unusual setup to trigger: two numbered exchanges sharing one anchor (`duplicate exchange r1-2`); `xref`/`xcomment` pointing at a label or anchor that doesn't exist anywhere (`xref(<fig-x>): not found`, `xcomment(<r9-9>): no exchange found for this anchor`); and an excerpt whose passage contains a label also referenced elsewhere in the bundle, which falls back to citing the page instead of crashing the compile outright — rare in practice, since `pinpoint(excerpt: true)` already strips labels from what it re-emits before this check would even trigger.

`set-strict(true)`, or `revisions(strict: true, ...)`, turns every one of these into a hard compile error, in both modes:

```
#set-strict(true)
```

Not the default, since a manuscript mid-revision should still compile — meant for a CI gate right before submission, when every one of these should already be resolved.

11. `revisions`: the pilot, and the final assembly

```
#show: revisions.with(template: ..., exchanges: ..., letter-template: auto, letter: auto, strict:
false)
```

Every example so far ran `add`/`del`/`passage`/`change-list`/… directly, no bundle. `revisions` is the one piece that actually needs one — it's what turns a manuscript and a set of exchanges into the three files of a real project.

A project is normally three files:

```
manuscript.typ    the manuscript, annotated with passage()/add()/del()/...
responses.typ      the exchanges with reviewers
main.typ           the pilot (a handful of lines)
```

`main.typ`:

```
#import "../../lib.typ": *

#let my-template(title: none, authors: (), body) = {
  set page(width: 16.6cm, height: auto, margin: 12pt)
  set text(size: 10.5pt)
  set heading(numbering: "1.")
  align(center, text(size: 1.3em, weight: "bold")[#title])
  v(0.5em)
  body
}

#let my-letter-template(body) = {
  set page(width: 16.6cm, height: auto, margin: 12pt)
  set text(size: 10.5pt)
  default-letter-template(body)
}

#show: revisions.with(
  template: my-template.with(title: [A Minimal Study]),
  letter-template: my-letter-template,
  exchanges: include "shared/responses.typ",
)

#include "shared/manuscript.typ"
```

`manuscript.typ`:

```
#import "../../../lib.typ": *

= Introduction

#added(<r1-1>)[The study rationale, expanded per the reviewer's request.]

#touched(<r1-2>)[This paragraph is unchanged; addressed in our response instead.]
```

`responses.typ`:

```
#import "../../../lib.typ": *

#reviewer(1)[
  #exchange(<r1-1>)[Please expand the rationale.][
    Done. #pinpoint(<r1-1>)
  ]

  #exchange(<r1-2>)[This point seems underdeveloped.][
    We believe the current wording is sufficient. #pinpoint(<r1-2>)
  ]
]
```

Two commands, run one after the other, produce the whole project:

```
typst compile --features bundle --format bundle main.typ
typst compile --features bundle --format bundle --input mode=tracked main.typ
```

First command (mode defaults to clean) — `manuscript.pdf`:

> **A Minimal Study**
>
> **1. Introduction**
>
> The study rationale, expanded per the reviewer's request.
>
> This paragraph is unchanged; addressed in our response instead.

— and `response.pdf`:

> **Response to Reviewers**
>
> **Reviewer 1**
>
> **Reviewer 1 — comment 1**
> *Please expand the rationale.*
>
> Done. (modified on p. 1)
>
> **Reviewer 1 — comment 2**
> *This point seems underdeveloped.*
>
> We believe the current wording is sufficient. (see p. 1)

Second command (`--input mode=tracked`) — `manuscript-tracked.pdf`, and **only** that file: this second compile never produces `response.pdf` again, and the first never produces a tracked manuscript — each compile produces exactly the file(s) for its own mode, never a mix of the two:

> **A Minimal Study**
>
> **1. Introduction**
>
> The study rationale, expanded per the reviewer's request. [R1-1]
>
> This paragraph is unchanged; addressed in our response instead. [R1-2]

Five parameters:

`template` wraps the manuscript only — any `content -> content` function, including a real journal template used the normal way. `authors:`/`title:` above are this example's own stand-in template's parameters, not `revisions`'s.

`letter-template` wraps the letter only, separately — `auto` (the default) uses `default-letter-template`, a title and nothing else. The two templates never mix: a figure/table/heading style set inside one has no effect on the other.

`exchanges` the already-evaluated content of the responses file — `include "responses.typ"`, as shown above.

`letter` whether this compile produces a response document at all — see below.

`strict` `revisions(strict: true, ...)` is shorthand for `set-strict(true)` at the top of the pilot.

Each source file needs its own `#import "palimpsest/lib.typ": *` — `#include` doesn't share scope, so `manuscript.typ` and `responses.typ` both import it too, even though only `main.typ` calls `revisions`.

11.1. `letter:` — matching the letter to the manuscript you actually send

This option changes nothing about the command line — it's still the exact same two commands shown above. Only the `letter: ...` value passed to `revisions()` inside `main.typ` changes what each of those two commands writes to disk.

By default, `response.pdf` is only ever produced by the clean compile, and its `pinpoint/xref` calls resolve against `manuscript.pdf` — even if the person reading the letter is actually looking at `manuscript-tracked.pdf`. `letter: true` produces a response document in **both** compiles, so the tracked one can quote the tracked wording and cite the tracked manuscript's own pagination.

```
#import "../../lib.typ": *

#let my-template(body) = {
  set page(width: 16.6cm, height: auto, margin: 12pt)
  set text(size: 10.5pt)
  body
}

#let my-letter-template(body) = {
  set page(width: 16.6cm, height: auto, margin: 12pt)
  set text(size: 10.5pt)
  default-letter-template(body)
}

#let exchanges = reviewer(1)[
  #exchange(<r1-1>)[This claim needs more support.][
    We have expanded this sentence. #pinpoint(<r1-1>, excerpt: true)
  ]
]

#show: revisions.with(
  template: my-template,
  letter-template: my-letter-template,
  exchanges: exchanges,
  letter: true,
)

#passage(<r1-1>)[The treatment #rep[has an effect][has a clinically meaningful effect] on
survival.]
```

Compiled clean, `response.pdf` quotes the accepted wording only:

Response to Reviewers

Reviewer 1

Reviewer 1 — comment 1

This claim needs more support.

We have expanded this sentence. **p. 1** — The treatment has a clinically meaningful effect on survival.

Compiled with `--input mode=tracked`, `letter: true` additionally produces `response-tracked.pdf` — a name distinct from `response.pdf`, so neither compile can silently overwrite the other's output. Its excerpt shows the struck-through old wording next to the underlined new wording, because `pinpoint(excerpt: true)` with no explicit `mode:` always follows whichever mode the **current** compile is running under (see pinpoint above):

Response to Reviewers

Reviewer 1

Reviewer 1 — comment 1

This claim needs more support.

We have expanded this sentence. **p. 1** — The treatment ~~has an effect~~ has a clinically meaningful effect on survival.

Three settings:

`auto` **(default)** unchanged from every earlier example in this manual — a letter only in the clean compile, as `response.pdf`. The tracked compile produces `manuscript-tracked.pdf` alone.

`true` a letter in both compiles, the tracked one named `response-tracked.pdf`.

`false` no letter in **either** compile — for a pure change-tracking project that never writes a response document, where `auto` would otherwise still produce a near-empty `response.pdf` from the clean compile.

What each command writes, for each setting:

Command	`letter: auto`	`letter: true`	`letter: false`
1st (clean)	`manuscript.pdf` `response.pdf`	`manuscript.pdf` `response.pdf`	`manuscript.pdf`
2nd (tracked)	`manuscript-tracked.pdf`	`manuscript-tracked.pdf` `response-tracked.pdf`	`manuscript-tracked.pdf`